



ALEXANDRIA UNIVERSITY

Faculty of Engineering
Computer and Communications Department

AI-Based Intelligent Surveillance System

Face Recognition and Audio Analysis for Public Security

Supervised By:

Dr. Mohamed Salah

Examined By:

Dr. Mohamed ElShimy

Team Members:

Mohamed Sobhy

Zyad Tarek Ahmed

Zeiad Emad

Youssef Emad

Berdj Youssef George

Academic Year: 2025 – 2026

Contents

Acknowledgment	8
Abstract	9
I Introduction and Background	10
1 Introduction	11
1.1 Introduction	11
1.2 Project Vision	11
1.3 Project Mission	11
1.4 Problem Statement	12
1.5 Proposed Solution	13
1.6 Scope of the Project	13
1.7 Report Structure	13
1.8 Summary	13
2 Literature Survey	14
2.1 Introduction	14
2.2 Face Detection and Recognition	14
2.3 Demographic Attribute Estimation	14
2.4 Emotion Recognition	14
2.5 Violence Detection in Video	15
2.6 Audio Analysis for Surveillance	15
2.7 Gaps Addressed by This Work	15
2.8 Summary	15
II Visual Analysis Modules	16
3 System Overview and Architecture	17
3.1 Introduction	17
3.2 System Architecture	17
3.2.1 Camera Module	17
3.2.2 Face Detection Module	17
3.2.3 Feature Extraction Module	17
3.2.4 Face Recognition Module	18
3.2.5 Demographic Estimation Module	18
3.2.6 Database Module	18

3.2.7	Security Assistance Module	18
3.2.8	Audio Processing Module	18
3.3	Core Modules Performance Summary	18
3.4	Data Flow Overview	19
3.5	Summary	19
4	Face Detection, Ethnicity, and Gender Detection	20
4.1	Introduction	20
4.2	Face Detection Module	20
4.2.1	Technical Implementation	20
4.2.2	Preprocessing Pipeline	20
4.2.3	Integration Points	21
4.3	Ethnicity Detection Module	21
4.3.1	Classification Categories	21
4.3.2	Dataset: FairFace	21
4.3.3	Model Architecture	21
4.3.4	Results	21
4.4	Gender Classification Module	22
4.4.1	Model: FairFace ONNX	22
4.4.2	Integration Approach	22
4.4.3	Applications	22
4.5	Summary	22
5	Age Estimation and Emotion Detection	23
5.1	Introduction	23
5.2	Age Estimation Module	23
5.2.1	Classification Categories	23
5.2.2	Model Architecture: MobileNetV2	23
5.2.3	Dataset	23
5.2.4	Results	24
5.3	Emotion Detection Module	24
5.3.1	Classification Categories	24
5.3.2	Dataset	24
5.3.3	Model Comparison	24
5.3.4	Final Model: EfficientNet	24
5.3.5	Integration and Applications	25
5.3.6	Challenges Faced	25
5.4	Summary	25
6	Violence Detection and Accessories Analysis	26
6.1	Introduction	26
6.2	Violence Detection Module	26
6.2.1	Dataset: RWF-2000	26
6.2.2	Methodology	26
6.2.3	Results and Applications	27
6.2.4	Challenges Addressed	27
6.3	Glasses Detection Module	27
6.3.1	Dataset	27
6.3.2	Model Architecture and Training	27

6.4	Head Pose Estimation and Occlusion Analysis Module	27
6.4.1	Dataset	28
6.4.2	Model Architecture	28
6.4.3	Results	28
6.5	Facial Hair Detection Module	28
6.5.1	Dataset	28
6.5.2	Model Architecture and Training	28
6.6	Summary	28
7	Face Recognition and Identity Management	29
7.1	Introduction	29
7.2	Implementation Details	29
7.3	Demographic Attribute Integration	29
7.4	Retrospective Identification Use Case	29
7.5	Privacy-Aware Storage	30
7.6	Summary	30
III	Audio Processing Pipeline	31
8	Audio Pipeline and Speech Recognition	32
8.1	Introduction	32
8.2	Audio Processing Pipeline Architecture	33
8.2.1	Advantages of the Proposed Architecture	33
8.3	Audio Acquisition and Capture	33
8.3.1	Fixed-Duration Recording	33
8.3.2	Silence-Based Recording	34
8.4	Audio Signal Preprocessing	35
8.4.1	Resampling	35
8.4.2	Mono Channel Conversion	35
8.4.3	Temporary Audio Storage	35
8.5	Speaker Recognition Using ECAPA-TDNN	35
8.5.1	ECAPA-TDNN Architecture	35
8.6	Speaker Enrollment and Data Augmentation	36
8.6.1	Speaker Enrollment Process	36
8.6.2	Audio Augmentation Strategy	36
8.7	Speaker Identification Using Cosine Similarity	37
8.7.1	Dynamic Thresholding	37
8.7.2	Identification Decision	37
8.8	Speech-to-Text Using Whisper Large-v3	38
8.8.1	Why Whisper Large-v3?	38
8.8.2	Language Detection	38
8.8.3	Confidence Estimation	38
8.8.4	GPU Acceleration	39
8.9	Mathematical Formulation of the Audio Pipeline	39
8.10	Azure Speech Backend (Alternative)	40
8.11	Summary	40

9	Language Detection and Real-Time Translation	41
9.1	Introduction	41
9.2	Language Processing Architecture	41
9.2.1	Advantages of the Proposed Framework	42
9.3	Hierarchical Language Detection Framework	42
9.4	Unicode Script Heuristic	43
9.4.1	Benefits of Script-Based Detection	43
9.5	Whisper-Based Language Detection	43
9.5.1	Confidence Threshold Strategy	44
9.6	Text-Based Ensemble Classification	44
9.6.1	LangDetect Engine	44
9.6.2	LangID Engine	44
9.6.3	Trigram Language Heuristics	44
9.6.4	Weighted Soft Voting	44
9.7	Confidence Blending and Temporal Smoothing	45
9.8	Real-Time Translation Using MarianMT	45
9.8.1	Architecture: MarianMT + Google Fallback	45
9.8.2	Supported MarianMT Language Pairs	46
9.8.3	Language Code Normalization	46
9.8.4	Passthrough Mode	46
9.9	Right-to-Left Language Handling	46
9.10	Integration with the Full Pipeline	46
9.11	Mathematical Formulation Summary	47
9.12	Summary	47
10	Violence Keyword Detection and Alert System	48
10.1	Introduction	48
10.2	Violence Detection System Architecture	48
10.2.1	Advantages of the Proposed Design	49
10.3	Multilingual Violence Detection Framework	49
10.4	Language-Aware Keyword Selection	49
10.5	Threat Detection Methodology	50
10.5.1	Latin Script Languages	50
10.5.2	Non-Latin Script Languages	50
10.6	Threat Detection Workflow	50
10.7	Email-Based Security Alert System	51
10.7.1	Alert Trigger	51
10.7.2	SMTP Configuration	51
10.8	HTML Alert Formatting	51
10.9	Right-to-Left Language Support	52
10.10	Full Pipeline Integration	52
10.11	Command-Line Interface	52
10.12	Example Pipeline Output	53
10.13	Mathematical Formulation Summary	53
10.14	Limitations and Future Improvements	54
10.15	Summary	54

IV Database, Ethics, and Conclusion	55
11 Database and System Integration	56
11.1 Introduction	56
11.2 Data Storage Strategy	56
11.3 Similarity Search	57
11.4 Backward Identification	57
11.5 Retrospective Identification Use Case	57
11.6 System Integration	57
11.7 Summary	57
12 Ethical Considerations	58
12.1 Introduction	58
12.2 Purpose Limitation	58
12.3 Human-in-the-Loop Design	58
12.4 Use of Soft Biometrics	58
12.5 Privacy Preservation	58
12.6 Bias Awareness and Mitigation	59
12.6.1 Audio Bias	59
12.6.2 Ethical Use in Security Applications	59
12.7 Legal Compliance	59
12.8 Ethical Compliance Summary	59
12.9 Summary	60
13 Conclusion and Future Work	61
13.1 Introduction	61
13.2 Summary of Accomplishments	61
13.3 Lessons Learned	62
13.4 Future Work	62
13.4.1 Visual Modules	62
13.4.2 Audio Module	63
13.4.3 System Integration	63
13.5 Final Remarks	63
References	64

List of Figures

List of Tables

3.1	Classification Categories and Performance by Module	19
5.1	Model Architecture Comparison for Emotion Detection	24
8.1	Fixed-Duration Recording Parameters	34
8.2	ECAPA-TDNN Speaker Recognition Configuration	38
8.3	Whisper large-v3 Configuration Parameters	39
9.1	Unicode Script to Language Mapping	43
9.2	Whisper Confidence Thresholds	44
9.3	Text Engine Weights in Soft Vote Mode	45
9.4	Selected Supported MarianMT Language Pairs	46
10.1	Supported Languages for Violence Keyword Detection	49
10.2	SMTP Alert Configuration	51

Acknowledgment

We would like to express our sincere gratitude to our project supervisor **Dr. Mohamed Salah** for his continuous guidance, valuable feedback, and unwavering support throughout the development of this project. His expertise and encouragement played a vital role in the successful completion of this work.

We also extend our appreciation to **Dr. Mohamed ElShimy** for his time, insightful comments, and constructive evaluation, which contributed greatly to improving the quality and clarity of this report.

Finally, we would like to thank our families and friends for their continuous encouragement and understanding throughout this journey.

Abstract

This graduation project presents the development of an **AI-Based Intelligent Surveillance System** designed to enhance security operations and demographic analysis in public spaces such as shopping malls, parks, and commercial centers. The system leverages computer vision and deep learning techniques to automatically analyze facial attributes, detect behavioral patterns, and provide actionable insights for security personnel and business intelligence.

The system architecture comprises multiple specialized modules working in concert: face detection and recognition for visitor identification, demographic attribute estimation (age, gender, ethnicity), emotional state analysis, behavioral pattern recognition (violence detection), and a comprehensive audio processing pipeline for situational awareness. Each module has been developed, trained, and validated independently to ensure modularity and ease of integration.

Key technical achievements include ethnicity detection at 82.56% accuracy using ResNet-18, emotion recognition at 73% accuracy with EfficientNet, age estimation at 77.23% using MobileNetV2, and gender classification at 99% accuracy with FairFace. The audio module implements a Whisper-first hierarchical language classification pipeline, multilingual violence keyword detection spanning over 30 languages, real-time speaker recognition using ECAPA-TDNN, and MarianMT-powered translation—all integrated into a single processing pipeline with automated email alerting.

The system is designed with ethical considerations at its core, operating as a human-in-the-loop decision support tool rather than an autonomous judgment system.

Part I

Introduction and Background

Chapter 1

Introduction

1.1 Introduction

In this chapter, we provide a comprehensive overview of the motivations and goals behind the **AI-Based Intelligent Surveillance System**. We begin by introducing the challenges faced in modern public security, followed by the vision and mission that drove our team to build this system. We then outline the project scope, the technologies adopted, and the ethical framework that governs its use.

1.2 Project Vision

The vision of this project is to create a comprehensive AI-powered surveillance system that transforms how public spaces handle security, safety, and operational intelligence.

Imagine a bustling shopping mall where a distraught mother approaches security, reporting that she has lost her child. Instead of manually reviewing hours of surveillance footage, the security officer simply inputs a description: “my son, about 8 years old, wearing a red shirt, Asian.” Within seconds, our intelligent system searches through live camera feeds and recorded footage, identifying potential matches and tracking the child’s movement across the mall. The system highlights the child’s last known location, enabling security personnel to respond swiftly and reunite the family.

Beyond commercial applications, our system serves critical security functions. Law enforcement agencies can integrate the system with drones to search for missing persons or criminal suspects in crowded public spaces such as parks, stadiums, or transportation hubs. The system processes live video streams, identifying individuals matching specific descriptions while maintaining ethical oversight and human decision-making authority.

This vision encompasses a future where AI enhances human capabilities rather than replacing human judgment, where technology serves communities by making public spaces safer and more responsive, and where ethical considerations remain at the forefront of every implementation decision.

1.3 Project Mission

Our mission has been to develop and validate the core AI modules that make this vision achievable. Throughout this project, we have successfully implemented and tested

multiple specialized detection and classification modules that form the foundation of the comprehensive surveillance system.

Specifically, our accomplishments include:

- Developed and trained a robust ethnicity detection module achieving 82.56% accuracy across six demographic categories.
- Implemented an emotion detection system capable of identifying seven emotional states with 73% accuracy.
- Created a violence detection module for analyzing video sequences to identify aggressive behavior in real-time.
- Built an age estimation module using MobileNetV2 architecture achieving 77.23% accuracy across nine age ranges.
- Integrated gender classification with 99% accuracy using the FairFace model.
- Developed a comprehensive audio processing pipeline supporting real-time speech recognition, language detection, multilingual violence keyword scanning, and automated alert dispatch.
- Designed and implemented a glasses detection module using ResNet-50 with transfer learning.
- Established a modular architecture allowing independent module development and future integration.
- Implemented ethical safeguards ensuring the system operates as a decision-support tool with human oversight.

1.4 Problem Statement

Traditional surveillance systems rely heavily on human operators to monitor camera feeds in real time. This approach is prone to fatigue, limited in scalability, and ineffective when handling large camera networks. Key challenges include:

- **Human Limitation:** Security officers cannot continuously monitor dozens of feeds simultaneously without missing critical events.
- **Retrospective Investigation:** Reviewing hours of recorded footage to find a suspect is time-consuming and resource-intensive.
- **Multilingual Environments:** Malls and public spaces host visitors speaking diverse languages, making audio monitoring without automated translation impractical.
- **Scalability:** Manual systems do not scale as the number of cameras or monitored spaces grows.

1.5 Proposed Solution

The proposed system addresses these limitations through a modular AI pipeline that automates the detection, classification, and alerting processes. The system is designed to assist—not replace—human security personnel, providing them with timely, accurate, and actionable information.

1.6 Scope of the Project

The system is scoped for deployment in semi-public and public spaces such as:

- Shopping malls and commercial centers
- Airports and transportation hubs
- University campuses and institutional buildings
- Parks and public event venues

The project covers both visual and audio analysis, with all modules developed to operate independently and integrate seamlessly through standardized interfaces.

1.7 Report Structure

The remainder of this report is organized as follows. Chapter 2 presents the literature survey. Chapter 3 provides the system overview and architecture. Chapters 4 through 9 describe the visual analysis modules (face detection, ethnicity, gender, age, emotion, violence, accessories, and face recognition). Chapters 10 through 12 detail the audio processing pipeline. Chapter 13 covers the database design, ethical framework, and conclusion.

1.8 Summary

In this chapter, we introduced the AI-Based Intelligent Surveillance System, outlined the motivations and challenges driving its development, and described the scope of the project. In the next chapter, we will survey the relevant literature and related work that informed our design decisions.

Chapter 2

Literature Survey

2.1 Introduction

This chapter surveys existing work in the areas of face detection and recognition, demographic attribute estimation, behavioral analysis, and audio processing for surveillance applications. We review state-of-the-art methods and datasets, and identify gaps that our system addresses.

2.2 Face Detection and Recognition

Face detection has evolved significantly from classical approaches such as Viola-Jones to modern deep learning detectors. The Multi-task Cascaded Convolutional Network (MTCNN) introduced a three-stage pipeline that jointly detects faces and localizes facial landmarks, offering high accuracy across pose and illumination variations. Deep face recognition models such as FaceNet, ArcFace, and the `face_recognition` library (based on dlib’s ResNet-34) produce compact 128-dimensional embeddings that enable efficient similarity search in large databases.

2.3 Demographic Attribute Estimation

Early demographic estimation relied on hand-crafted features and shallow classifiers. Deep learning transformed the field: ResNet and EfficientNet architectures now achieve near-human accuracy on age, gender, and ethnicity estimation benchmarks. The Fair-Face dataset was specifically designed to address demographic bias present in earlier datasets such as CelebA, providing balanced representation across seven racial groups. MobileNetV2’s inverted residual structure makes it particularly suitable for age-range classification tasks where lightweight real-time inference is required.

2.4 Emotion Recognition

Facial action coding systems provided early frameworks for emotion analysis. Convolutional networks trained on datasets such as AffectNet and FER2013 have since achieved

strong performance across the standard seven emotion categories. EfficientNet’s compound scaling offers an effective balance between accuracy and computational cost, making it a popular choice for real-time emotion analysis in surveillance contexts.

2.5 Violence Detection in Video

Video-based violence detection requires temporal modeling beyond single-frame analysis. The RWF-2000 dataset provides 2000 real-world surveillance clips balanced between violent and non-violent scenes. Optical flow features combined with CNN-LSTM or 3D-CNN architectures have demonstrated strong performance on this benchmark. Frame-level spatial features combined with temporal aggregation enable the distinction between legitimate physical activity and actual violence.

2.6 Audio Analysis for Surveillance

Automatic speech recognition (ASR) systems have advanced dramatically with transformer-based architectures. OpenAI Whisper, trained on 680,000 hours of multilingual audio, achieves near-human transcription accuracy across 99 languages and provides built-in language detection through spectral analysis. Speaker recognition systems based on ECAPA-TDNN (Emphasized Channel Attention, Propagation, and Aggregation in TDNN) achieve state-of-the-art performance on the VoxCeleb benchmark. For translation, MarianMT provides efficient neural machine translation across hundreds of language pairs, with Google Translate serving as a reliable fallback for unsupported pairs.

2.7 Gaps Addressed by This Work

Existing systems typically address either visual or audio surveillance in isolation. Our system bridges this gap by integrating both modalities into a unified pipeline. Furthermore, most audio surveillance work focuses on English; our multilingual violence keyword detection spans over 30 languages, making the system viable in diverse cultural settings.

2.8 Summary

This chapter surveyed the key technologies and datasets that underpin our system. In the next chapter, we present the overall system architecture and describe how the individual modules interact.

Part II

Visual Analysis Modules

Chapter 3

System Overview and Architecture

3.1 Introduction

This chapter presents the overall architecture of the AI-Based Intelligent Surveillance System. We describe how the individual modules are organized, how data flows between them, and how the system integrates visual and audio analysis into a unified pipeline.

3.2 System Architecture

The proposed intelligent surveillance system consists of seven main visual components and one comprehensive audio component, all working together to provide analysis and decision support. Each component is designed as a modular unit that can be developed, tested, and deployed independently while maintaining seamless integration through standardized interfaces.

3.2.1 Camera Module

Captures real-time video streams from cameras installed in strategic locations such as mall entrances, shop interiors, and public areas. The module supports multiple camera inputs and provides frame-level extraction for downstream processing.

3.2.2 Face Detection Module

Detects faces within video frames using deep learning-based face detectors. This module serves as the entry point for all facial analysis, identifying and cropping face regions for feature extraction while handling variations in lighting, pose, and occlusion.

3.2.3 Feature Extraction Module

Extracts unique facial embeddings (128-dimensional numerical feature vectors) that represent each detected face. These embeddings are invariant to minor changes in lighting, pose, and facial expression, enabling reliable comparison across different captures of the same person.

3.2.4 Face Recognition Module

Compares extracted embeddings with stored embeddings in the database to determine whether the person already exists in the system. This enables visitor re-identification and tracking across multiple visits or camera locations.

3.2.5 Demographic Estimation Module

Estimates soft biometric attributes including age range, gender, ethnicity, emotional state, and appearance-based features using specialized neural networks trained for each classification task.

3.2.6 Database Module

Stores facial embeddings, timestamps, visit history, and estimated attributes in a secure database structure. The database supports efficient similarity searches and filtering operations for retrospective analysis.

3.2.7 Security Assistance Module

Allows retrospective search and filtering based on demographic attributes and visit history to assist security personnel during investigations. The module provides candidate shortlists rather than definitive identifications, maintaining human oversight.

3.2.8 Audio Processing Module

Captures ambient audio, transcribes speech using Whisper large-v3, detects the spoken language, translates to a target language, identifies the speaker using ECAPA-TDNN, scans the transcription for violence keywords in over 30 languages, and dispatches automated email alerts when threats are detected.

3.3 Core Modules Performance Summary

Table 3.1 summarizes the classification categories and accuracy achieved by each visual module.

Module	Categories	Accuracy
Ethnicity	White, Black, Middle Eastern, Indian, Asian, Latino	82.56%
Emotion	Neutral, Happy, Sad, Fear, Angry, Surprise, Disgust	73%
Age Range	0–3, 4–12, 13–17, 18–25, 26–35, 36–45, 46–55, 56–70, 70+	77.23%
Gender	Male, Female	99%
Glasses	With Glasses, Without Glasses	90%
Head Pose	Yaw, Pitch, Roll angles	MAE $\approx 3.8^\circ$
Occlusion	Occlusion level	MAE ≈ 0.07

Table 3.1: Classification Categories and Performance by Module

3.4 Data Flow Overview

The end-to-end data flow is as follows:

1. Camera frames are captured and forwarded to the Face Detection Module.
2. Detected face regions are cropped and normalized to 224×224 pixels.
3. The Feature Extraction Module generates a 128-dimensional embedding per face.
4. The Face Recognition Module compares the embedding against the database.
5. All demographic modules (age, gender, ethnicity, emotion, glasses, head pose, facial hair) run in parallel on the cropped face.
6. Results are stored in the database alongside the embedding and timestamp.
7. Concurrently, the Audio Processing Module captures ambient audio, transcribes it, detects language, translates, identifies the speaker, and scans for violence keywords.
8. If a threat keyword is detected, an automated alert email is dispatched to the designated recipient.
9. The Security Assistance Module enables retrospective queries combining visual and audio evidence.

3.5 Summary

This chapter described the overall architecture and data flow of the system. The following chapters detail each visual module. The audio pipeline is covered in Chapters 10 through 12.

Chapter 4

Face Detection, Ethnicity, and Gender Detection

4.1 Introduction

This chapter covers three foundational modules: face detection (which feeds all other visual modules), ethnicity classification, and gender classification. Together, these modules form the primary demographic profiling layer of the surveillance system.

4.2 Face Detection Module

The face detection module serves as the foundation for all facial analysis in the system. It processes input video frames or images to identify and localize human faces, outputting bounding box coordinates for each detected face.

4.2.1 Technical Implementation

The module utilizes the Multi-task Cascaded Convolutional Networks (MTCNN) framework, which provides a three-stage architecture that progressively refines face detection and landmark localization:

- **Stage 1 (Proposal Network):** Generates initial face candidates through a shallow CNN.
- **Stage 2 (Refine Network):** Filters false positives and refines bounding boxes.
- **Stage 3 (Output Network):** Produces final detections with facial landmark positions.

4.2.2 Preprocessing Pipeline

Detected faces undergo preprocessing to ensure consistent input for classification modules:

- Face alignment based on eye positions to normalize head pose.
- Cropping to square aspect ratio around the face center.

- Resizing to 224×224 pixels for standard model input.
- Normalization using ImageNet statistics for pre-trained model compatibility.

4.2.3 Integration Points

The face detection module interfaces with all facial analysis modules, providing cropped face images to ethnicity, emotion, age, gender, and glasses detection components. It also generates facial embeddings for the recognition module and landmark points for auxiliary analysis.

4.3 Ethnicity Detection Module

The ethnicity detection module classifies a person's ethnicity based on facial features captured from surveillance cameras. This information assists in person profiling, suspect tracking, and search-space reduction during incident investigations.

4.3.1 Classification Categories

The system classifies faces into six categories:

- White • Black • Middle Eastern
- Indian • Asian • Latino

4.3.2 Dataset: FairFace

The FairFace dataset was selected for training due to its balanced representation across different races, genders, and age groups. The dataset contains approximately 108,000 facial images with high-quality annotations designed to reduce bias in facial attribute classification.

4.3.3 Model Architecture

Two architectures were evaluated:

- **Basic CNN:** A custom-designed convolutional neural network served as the baseline. While achieving reasonable performance, it showed limitations in generalization for challenging facial images.
- **ResNet-18 (Selected):** A deep residual network with skip connections. ResNet-18, pre-trained on ImageNet and fine-tuned on FairFace, outperformed the basic CNN due to its deeper architecture capable of learning complex facial features, residual connections that improve training stability, and better generalization on unseen data.

4.3.4 Results

The final ethnicity detection module achieves **82.56%** accuracy on test data. The module outputs classification probabilities for each category, enabling confidence-based filtering during retrospective searches.

4.4 Gender Classification Module

The gender classification module provides binary gender classification (Male/Female) with high accuracy, serving as a fundamental demographic attribute for visitor analytics and retrospective search filtering.

4.4.1 Model: FairFace ONNX

The gender classification module utilizes the FairFace model, achieving **99%** accuracy in binary gender classification. The model is deployed using ONNX runtime for optimized inference performance.

4.4.2 Integration Approach

The module integrates seamlessly with the face detection pipeline, receiving cropped face images and outputting gender predictions stored in the database alongside other demographic attributes.

4.4.3 Applications

Gender classification supports demographic analytics, targeted marketing analysis, and provides an additional filtering attribute for security investigations.

4.5 Summary

This chapter described the face detection pipeline, the ethnicity detection module achieving 82.56% accuracy, and the gender classification module achieving 99% accuracy. In the next chapter, we cover age estimation and emotion detection.

Chapter 5

Age Estimation and Emotion Detection

5.1 Introduction

This chapter covers two modules that contribute to a deeper understanding of individuals captured by the surveillance system: age estimation and emotion detection. Both modules operate on cropped face images produced by the face detection pipeline.

5.2 Age Estimation Module

The age estimation module predicts a person’s age range from facial images. Rather than predicting exact numerical ages, which is prone to high error rates, the system classifies individuals into predefined age categories. This approach provides more reliable and interpretable results suitable for demographic analysis.

5.2.1 Classification Categories

The module classifies faces into nine age ranges:

0–3 4–12 13–17 18–25 26–35 36–45 46–55 56–70 70+

5.2.2 Model Architecture: MobileNetV2

Age estimation was implemented as an age-range classification problem using a MobileNetV2-based convolutional neural network. Several architectures were initially evaluated, including a basic CNN and alternative lightweight MobileNet variants. However, these models showed lower accuracy and weaker generalization, particularly under varying lighting conditions and facial expressions.

MobileNetV2 was selected as the final model due to its superior performance-efficiency trade-off, stable convergence during training, and suitability for real-time applications. The inverted residual structure with linear bottlenecks enables efficient feature extraction while maintaining accuracy comparable to heavier models.

5.2.3 Dataset

The model was trained on a manually collected facial image dataset from Kaggle containing approximately 20,000 images with age annotations. The dataset provides balanced

representation across age ranges with sufficient samples for robust training within each category.

5.2.4 Results

The age estimation module achieves **77.23%** accuracy on test data. The predicted age range is determined by selecting the class with the highest softmax probability, providing a robust and interpretable estimate rather than an exact numerical age.

5.3 Emotion Detection Module

The emotion detection module identifies a person's emotional state from facial expressions. This capability is particularly valuable in surveillance scenarios for detecting fear in lost children, identifying anger or aggression before escalation, and enhancing overall situational awareness for security personnel.

5.3.1 Classification Categories

The module identifies seven emotional states: Neutral, Happy, Sad, Fear, Angry, Surprise, and Disgust.

5.3.2 Dataset

The emotion detection model was trained using a self-collected dataset containing images representing different human emotions, with balanced representation across emotion categories.

5.3.3 Model Comparison

Multiple deep learning architectures were evaluated:

Model	Description	Accuracy
Basic CNN	Custom shallow CNN	65%
ResNet	Deep residual network	70%
MobileNet	Lightweight model optimized for speed	73%
EfficientNet	Scaled architecture with optimized depth	78% (Selected)

Table 5.1: Model Architecture Comparison for Emotion Detection

5.3.4 Final Model: EfficientNet

EfficientNet achieved the best performance. The model was selected due to its balanced scaling of depth, width, and resolution, higher accuracy with fewer parameters compared to deeper architectures, and suitability for real-time surveillance systems where computational efficiency is critical. The system achieves **73%** accuracy on the held-out test set.

5.3.5 Integration and Applications

The emotion detection module is integrated with OpenCV for real-time emotion recognition from camera feeds. Applications include alerting security when individuals exhibit signs of distress, identifying potentially volatile situations based on aggressive expressions, and providing aggregated emotional analytics for space optimization.

5.3.6 Challenges Faced

- **Dataset Imbalance:** Some emotion categories (e.g., disgust, fear) are underrepresented in natural datasets, requiring careful augmentation.
- **Computational Constraints:** Training EfficientNet required significant GPU resources and long training times.
- **Real-Time Performance:** Achieving the balance between accuracy and inference speed was challenging for live deployment.
- **Lighting and Pose Variations:** Facial expressions are sensitive to illumination and head orientation changes.

5.4 Summary

This chapter described the age estimation module achieving 77.23% accuracy using MobileNetV2, and the emotion detection module achieving 73% accuracy using EfficientNet. In the next chapter, we cover violence detection and accessories detection.

Chapter 6

Violence Detection and Accessories Analysis

6.1 Introduction

This chapter covers the video-based violence detection module and three appearance-attribute modules: glasses detection, head pose estimation with occlusion analysis, and facial hair detection. These modules extend the system’s analytical capabilities beyond basic demographics.

6.2 Violence Detection Module

The violence detection module analyzes video sequences to automatically detect violent behavior. This feature is critical for early intervention in cases of physical fights, theft-related violence, and child kidnapping attempts.

6.2.1 Dataset: RWF-2000

The RWF-2000 dataset provides 2,000 real-world video clips balanced between violent and non-violent scenes. The dataset is captured from real surveillance cameras, ensuring that the trained model generalizes well to actual deployment environments.

6.2.2 Methodology

The violence detection pipeline processes video sequences through the following stages:

- Frame-by-frame processing to extract spatial features from individual images.
- Temporal feature extraction to capture motion patterns across consecutive frames.
- Sequence classification determining whether the video segment contains violent or non-violent behavior.

6.2.3 Results and Applications

The trained model successfully detects violence from video input and can handle real-world surveillance footage. The module forms a foundation for real-time alert systems that notify security personnel when potentially dangerous situations develop. Integration with the face recognition module enables identification of individuals involved in violent incidents.

6.2.4 Challenges Addressed

Real surveillance data contains complex scenes with varying lighting, camera angles, and crowd densities. The model must distinguish between legitimate physical activity (sports, dancing) and actual violence. Temporal modeling requires careful balance between capturing sufficient context and maintaining real-time performance.

6.3 Glasses Detection Module

The glasses detection module identifies whether an individual is wearing glasses, providing an important visual descriptor for person identification and reverse search queries.

6.3.1 Dataset

The module uses the CelebA dataset with over 200,000 celebrity images. A balanced sample of 500 images with glasses and 500 without was used for training. Faces were detected and cropped using MTCNN before training.

6.3.2 Model Architecture and Training

- **Architecture:** ResNet-50 (pre-trained on ImageNet) with skip connections for learning subtle features like thin frames.
- **Modifications:** Final layer replaced with a single output neuron with Sigmoid activation.
- **Input Size:** 224×224 pixels.
- **Optimizer:** Adam with learning rate 0.0001.
- **Loss Function:** BCEWithLogitsLoss for stable binary classification.
- **Performance:** 90% accuracy with reliable performance in live demonstrations.

6.4 Head Pose Estimation and Occlusion Analysis Module

This module estimates head position (yaw, pitch, roll angles) and measures facial occlusion level as a regression task predicting continuous values.

6.4.1 Dataset

The FFHQ (Flickr-Faces-HQ) dataset was used, providing high-quality face images with pose and occlusion annotations. Targets were normalized: pose divided by 90 (to 0–1 range), occlusion by 100.

6.4.2 Model Architecture

MobileNetV2 (pre-trained on ImageNet) serves as the backbone, with dual heads for pose (3 outputs) and occlusion (2 outputs). Global average pooling reduces features to a 1280-vector, followed by a shared 512-unit layer with dropout (0.3). Training used Huber Loss and Adam optimizer with partial unfreezing of layers.

6.4.3 Results

- Mean Absolute Error (MAE) for pose: $\approx 3.8^\circ$
- Pose MAE improved from 0.09 to 0.042 ($\approx 50\%$ improvement across trials).
- Occlusion MAE: ≈ 0.07

6.5 Facial Hair Detection Module

This module detects any facial hair (stubble, mustache, beard) as a binary classification task, generalizing beyond just beards for better detection of subtle growth.

6.5.1 Dataset

A custom “beard-or-no-beard” dataset with train, validation, and test folders was used. Labels are mapped as 1 for facial hair and 0 for none.

6.5.2 Model Architecture and Training

- **Architecture:** ResNet-50 (pre-trained on ImageNet) for deep texture detection.
- **Augmentation:** Horizontal flip, rotation (15°), color jitter.
- **Optimizer:** Adam with learning rate 0.0003 and weight decay 0.0001.
- **Loss Function:** BCEWithLogitsLoss.
- **Early Stopping:** Patience of 7 epochs.

6.6 Summary

This chapter covered the video-based violence detection module and three binary classification accessories modules: glasses detection (90%), head pose estimation (MAE $\approx 3.8^\circ$), and facial hair detection. In the next chapter, we describe the face recognition and identity management module.

Chapter 7

Face Recognition and Identity Management

7.1 Introduction

This chapter describes the face recognition and identity management module, which is responsible for capturing facial images from live video streams, encoding facial features, and identifying individuals by comparing them with existing records in the database.

7.2 Implementation Details

The module is implemented using OpenCV for video capture and visualization, combined with the `face_recognition` library for facial encoding and matching. Each detected face is converted into a fixed-length 128-dimensional numerical embedding that uniquely represents the person's facial features. Known individuals are pre-enrolled in the database with their embeddings and associated metadata.

7.3 Demographic Attribute Integration

The module is tightly integrated with demographic estimation components:

- **Age Estimation:** Performed using the trained MobileNetV2-based deep learning model, which classifies faces into predefined age ranges after preprocessing and normalization.
- **Gender Classification:** Performed using the FairFace ONNX model, integrated via OpenCV's DNN module for efficient and unbiased inference.

These attributes are overlaid on the video stream and stored in the database to support demographic analytics and forensic search.

7.4 Retrospective Identification Use Case

In the event of a reported incident such as theft, security personnel can provide a reference image of the suspect. The system encodes this image and performs a database-wide search

to identify matching or visually similar individuals. When an exact identity match is unavailable, demographic attributes (age range, gender, ethnicity) are used to narrow down potential candidates, significantly reducing investigation time.

7.5 Privacy-Aware Storage

The module stores facial data as numerical embeddings rather than raw facial images. Embeddings cannot be reverse-engineered to reconstruct the original face, providing privacy-aware storage while maintaining identification capability.

7.6 Summary

This chapter described the face recognition and identity management module, including demographic attribute integration and retrospective identification capabilities. The next three chapters detail the audio processing pipeline.

Part III

Audio Processing Pipeline

Chapter 8

Audio Pipeline and Speech Recognition

8.1 Introduction

Audio processing represents one of the fundamental components of the proposed AI-Based Intelligent Surveillance System because it enables the extraction of meaningful information from spoken interactions occurring in monitored environments. While conventional surveillance systems primarily rely on visual monitoring, modern intelligent surveillance requires the integration of audio intelligence to improve situational awareness, identify speakers, understand conversations, detect threats, and support multilingual environments.

The audio processing subsystem developed in this project was designed to provide a complete end-to-end pipeline for speech understanding and speaker identification. The implemented framework combines real-time audio acquisition, automatic speech recognition, speaker recognition, language-aware processing, and intelligent threat analysis into a unified architecture. The system was specifically optimized to operate efficiently in real-time surveillance scenarios while maintaining high transcription accuracy and robust speaker verification.

The proposed audio pipeline consists of five major stages:

- **Audio Capture**, responsible for recording audio signals from microphones.
- **Speaker Recognition**, which identifies the individual speaking using biometric voice embeddings.
- **Speech-to-Text (STT)**, which converts spoken audio into textual content.
- **Language Detection and Translation**, enabling multilingual support.
- **Violence Detection and Alerting**, which identifies dangerous speech content and generates security notifications.

The architecture was implemented using several state-of-the-art artificial intelligence frameworks including PyAudio, OpenAI Whisper large-v3, SpeechBrain ECAPA-TDNN, Transformers MarianMT, and additional language processing libraries. The integration of these technologies allows the system to function as an intelligent multilingual surveillance solution capable of identifying speakers and detecting potentially dangerous situations in real time.

8.2 Audio Processing Pipeline Architecture

The proposed audio processing system follows a hierarchical multi-stage architecture where each module performs a specialized task before passing its output to the next stage.

The implemented workflow begins by capturing audio either through fixed-duration recording or silence-aware recording. After audio acquisition, the speaker recognition module extracts voice embeddings and compares them against stored speaker profiles. The speech recognition module then converts speech into text using Whisper large-v3. Finally, higher-level semantic analysis modules process the transcribed content for language classification and violence detection.

The complete audio pipeline can be summarized as follows:

Audio Input → Speaker Identification → Speech Recognition → Language Analysis → Translation → Violence Detection → Alert Generation

The modular design ensures that each subsystem can function independently while maintaining interoperability with the complete surveillance framework. This architecture provides flexibility, scalability, and ease of maintenance.

8.2.1 Advantages of the Proposed Architecture

The proposed architecture offers several advantages:

- Real-time processing capability, enabling immediate speech analysis.
- High transcription accuracy through Whisper large-v3.
- Robust speaker recognition using ECAPA-TDNN embeddings.
- Multilingual support, allowing communication monitoring across languages.
- Threat awareness, enabling automatic violence detection.
- Scalable modular design, facilitating future extensions.

8.3 Audio Acquisition and Capture

Audio acquisition represents the first stage of the processing pipeline. The system captures sound signals using microphones connected to the surveillance platform. To achieve reliable performance in different environments, two recording modes were implemented.

8.3.1 Fixed-Duration Recording

In fixed-duration mode, audio is captured for a predefined duration. The implementation uses a default recording interval of five seconds, although the duration can be modified according to surveillance requirements.

The recorded signal is sampled using the following parameters:

Parameter	Value
Sampling Rate	16,000 Hz
Channels	1 (Mono)
Chunk Size	1024
Audio Format	16-bit PCM
Default Duration	5 Seconds

Table 8.1: Fixed-Duration Recording Parameters

The choice of a 16 kHz sampling frequency was motivated by the fact that speech information is primarily concentrated below 8 kHz, according to the Nyquist theorem. Therefore, a 16 kHz sampling rate provides sufficient quality for speech recognition while reducing computational cost.

The audio signal is continuously divided into chunks of size 1024 samples, which are sequentially buffered and stored in memory until recording completion. After recording, the chunks are merged and saved into a temporary WAV file for downstream processing.

8.3.2 Silence-Based Recording

In surveillance applications, fixed recording may capture unnecessary silence and increase computational overhead. To address this issue, a silence-aware recording mechanism was implemented.

Instead of recording for a fixed duration, the system continuously listens to environmental audio and automatically terminates recording when prolonged silence is detected.

The silence detection process relies on Root Mean Square (RMS) energy estimation. The RMS energy of an audio signal is computed as:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

where x_i represents the amplitude of each audio sample and N denotes the total number of samples.

When the RMS energy falls below a predefined silence threshold, the system considers the environment silent. A silence counter is incremented until a predefined silence duration is reached, after which recording stops automatically.

Benefits of Silence-Based Recording

The silence-based mechanism provides multiple benefits:

- Reduces unnecessary processing.
- Eliminates long silent recordings.
- Improves computational efficiency.
- Enables adaptive recording in real-world environments.

8.4 Audio Signal Preprocessing

Before higher-level analysis, the captured audio signal undergoes preprocessing to ensure consistency across different recording devices and acoustic conditions.

8.4.1 Resampling

Audio signals may originate from devices with varying sampling frequencies. Therefore, the system normalizes all audio signals to 16 kHz. Resampling ensures compatibility between input audio and machine learning models such as Whisper and ECAPA-TDNN, both of which were optimized for standardized speech representations.

8.4.2 Mono Channel Conversion

Multi-channel recordings can introduce inconsistencies during feature extraction. To overcome this limitation, stereo recordings are converted into a mono representation by averaging channels:

$$x_{\text{mono}} = \frac{x_{\text{left}} + x_{\text{right}}}{2}$$

This operation guarantees consistent audio representation during embedding generation.

8.4.3 Temporary Audio Storage

After preprocessing, the audio stream is stored temporarily in WAV format. This intermediate representation facilitates interoperability between different processing modules such as speaker recognition and speech recognition.

8.5 Speaker Recognition Using ECAPA-TDNN

Speaker recognition is a biometric authentication process used to identify individuals based on unique vocal characteristics. In surveillance systems, speaker identification enhances contextual understanding and improves forensic investigation capabilities.

The proposed system employs the ECAPA-TDNN (Emphasized Channel Attention, Propagation and Aggregation Time Delay Neural Network) architecture implemented through the SpeechBrain framework. The selected pretrained model was `speechbrain/spkrec-ecapa-v`, trained on the VoxCeleb dataset and demonstrating state-of-the-art performance in speaker verification tasks.

ECAPA-TDNN was selected because it offers high speaker verification accuracy, strong robustness against environmental noise, efficient embedding extraction, and excellent performance in real-world speech environments. Unlike traditional handcrafted voice features such as MFCCs, ECAPA-TDNN learns highly discriminative speaker representations automatically.

8.5.1 ECAPA-TDNN Architecture

The ECAPA-TDNN model generates high-dimensional speaker embeddings from input speech signals. These embeddings represent compact numerical vectors that encode speaker identity information. During enrollment, multiple embeddings are collected from

an individual speaker and averaged to form a speaker profile centroid. During inference, newly extracted embeddings are compared against stored centroids to determine identity.

The ECAPA-TDNN model improves upon traditional TDNN architectures through three major mechanisms:

Channel Attention Mechanism

Channel attention allows the model to focus on the most discriminative speaker-related information within speech features. This process improves robustness in noisy environments because irrelevant acoustic information receives lower importance during feature extraction.

Propagation Mechanism

Feature propagation enhances contextual learning by allowing deeper layers to access information from previous network layers. This improves feature representation and increases speaker discrimination performance.

Temporal Aggregation

Temporal aggregation combines speech information across time, allowing the model to capture long-term vocal patterns such as speaking rhythm, pitch variation, pronunciation style, and vocal resonance. These characteristics significantly improve speaker identification accuracy in surveillance applications.

8.6 Speaker Enrollment and Data Augmentation

Before speaker identification can be performed, the system requires an enrollment phase in which speaker voice profiles are created and stored.

8.6.1 Speaker Enrollment Process

During enrollment, the system accepts one or multiple speech recordings from an individual speaker. Each audio sample undergoes preprocessing and embedding extraction through ECAPA-TDNN. The extracted embeddings are stored and averaged to create a stable speaker representation called a speaker centroid:

$$C = \frac{1}{N} \sum_{i=1}^N E_i$$

where C represents the speaker centroid, E_i denotes the embedding vector extracted from sample i , and N represents the total number of recordings. The resulting centroid serves as the reference representation of the speaker.

8.6.2 Audio Augmentation Strategy

One challenge in speaker recognition systems is limited enrollment data. To improve robustness, the implemented system introduces automatic data augmentation during enrollment using two techniques:

Pitch Shifting

Pitch shifting artificially modifies vocal frequency while preserving speaker identity. Multiple pitch levels are generated: -2 semitones, -1 semitone, $+1$ semitone, and $+2$ semitones. This allows the system to generalize better to voice variations caused by mood, fatigue, or microphone conditions.

Time Stretching

Time stretching changes speech speed while preserving vocal pitch. The following stretch factors are used: $0.90\times$, $0.95\times$, $1.05\times$, and $1.10\times$. These augmentations improve system resilience against variations in speaking speed and significantly increase enrollment diversity.

8.7 Speaker Identification Using Cosine Similarity

After enrollment, speaker identification is performed by comparing incoming voice embeddings with stored speaker centroids using cosine similarity:

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

where A is the query embedding, B is the enrolled speaker centroid, $A \cdot B$ represents the dot product, and $\|A\|$ and $\|B\|$ represent vector magnitudes. The similarity value ranges from 1 (identical embeddings) to -1 (completely opposite representations).

8.7.1 Dynamic Thresholding

Instead of using a fixed threshold for all speakers, the proposed implementation employs speaker-specific thresholds dynamically computed during enrollment:

$$\text{Threshold} = \max(0.25, \mu - 2.5\sigma)$$

where μ represents the mean cosine similarity during enrollment and σ represents the standard deviation. This adaptive threshold improves recognition reliability because each speaker possesses unique vocal consistency.

8.7.2 Identification Decision

The speaker with the highest similarity score is selected. If the similarity exceeds the stored threshold, the speaker is accepted; otherwise, the result is reported as “Unknown Person.” This strategy minimizes false positive identification and prevents unauthorized speaker matching.

Parameter	Value
Pre-trained model	speechbrain/spkrec-ecapa-voxceleb
Embedding dimension	192
Similarity metric	Cosine similarity
Default global threshold	0.60
Augmentation	Pitch shift + time stretch
Storage format	Pickle (.pkl)

Table 8.2: ECAPA-TDNN Speaker Recognition Configuration

8.8 Speech-to-Text Using Whisper Large-v3

Speech recognition is responsible for converting spoken language into machine-readable text. The implemented system uses OpenAI Whisper large-v3, which is among the most accurate multilingual automatic speech recognition models currently available.

8.8.1 Why Whisper Large-v3?

Whisper large-v3 was selected due to several advantages:

- Exceptional multilingual transcription accuracy.
- Built-in language detection.
- Robustness against noisy environments.
- Support for low-resource languages.
- Strong performance in surveillance scenarios.

Unlike conventional speech recognition systems, Whisper directly learns from large-scale multilingual speech corpora, enabling generalization across accents and dialects.

8.8.2 Language Detection

A key advantage of Whisper is its ability to detect spoken language directly from acoustic signals. Instead of relying solely on transcribed text, Whisper analyzes audio spectrograms and returns a probability distribution across 99 supported languages. The system extracts the most probable language, its confidence score, and the top language probabilities. This language prediction is later integrated into the hierarchical language classification framework.

8.8.3 Confidence Estimation

The confidence of speech recognition is estimated using the average logarithmic probability generated across speech segments:

$$\text{Confidence} = e^{\text{AvgLogProb}}$$

where AvgLogProb represents the average log probability across transcription segments. Higher confidence values indicate more reliable speech recognition output.

8.8.4 GPU Acceleration

To improve computational performance, the system automatically detects available hardware. When a CUDA-enabled GPU is available, Whisper executes on the GPU; otherwise, CPU execution is used. This adaptive design improves portability and allows deployment on different hardware environments.

Parameter	Value
Model	Whisper large-v3
Beam Size	5
Best-of	5
Temperature	0
Word Timestamps	Enabled
No Speech Threshold	0.6

Table 8.3: Whisper large-v3 Configuration Parameters

8.9 Mathematical Formulation of the Audio Pipeline

The proposed audio processing pipeline can be mathematically summarized through its core components:

Audio Capture

The input audio signal:

$$x(t) = A \sin(2\pi ft + \phi)$$

where A is signal amplitude, f is frequency, and ϕ is phase.

RMS Energy (Silence Detection)

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

Speaker Embedding Similarity

$$\text{CosineSimilarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

Transcription Confidence

$$\text{Confidence} = e^{\text{AvgLogProb}}$$

These formulations collectively describe the mathematical foundations of the implemented audio processing framework.

8.10 Azure Speech Backend (Alternative)

For deployments requiring cloud-based transcription, the pipeline supports Microsoft Azure Cognitive Services Speech as an alternative backend. When selected, the system forwards the WAV file to the Azure Speech Recognition API and returns the transcribed text. This backend is particularly useful when on-device GPU resources are unavailable.

8.11 Summary

This chapter presented the design and implementation of the speech recognition and audio processing subsystem. The proposed architecture integrates audio acquisition, speaker recognition, and multilingual speech transcription into a unified processing pipeline. The system utilizes PyAudio for real-time audio capture, supports both fixed-duration and silence-based recording modes, and applies RMS-based silence detection for efficient audio acquisition. Speaker recognition is performed using ECAPA-TDNN, enabling robust biometric speaker identification through embedding extraction and cosine similarity matching. Furthermore, Whisper large-v3 was employed for multilingual speech recognition, providing highly accurate transcriptions together with language probability estimation. The implemented subsystem forms a critical foundation for subsequent stages of the surveillance system, including language detection, translation, and violence analysis, which are discussed in the following chapter.

Chapter 9

Language Detection and Real-Time Translation

9.1 Introduction

In intelligent surveillance systems, understanding spoken content is not limited to converting speech into text. Real-world environments often involve multilingual communication where individuals may speak different languages, dialects, or scripts. Therefore, an advanced surveillance framework must not only recognize speech but also determine the spoken language and translate it into a unified target language for effective monitoring.

Traditional language detection systems primarily depend on text analysis after transcription. However, text-based approaches frequently encounter ambiguity when dealing with short utterances, multilingual speech, or linguistically similar languages such as French and Dutch or Spanish and Portuguese. To overcome these limitations, the proposed surveillance system implements a hierarchical language detection framework that combines acoustic analysis, text-based classification, script detection, and confidence-based decision fusion.

The proposed implementation adopts a Whisper-first hierarchical classification strategy, where language identification begins with Unicode script analysis, followed by Whisper large-v3 acoustic language detection, and finally ensemble text classification using multiple language recognition engines. This layered architecture improves robustness and significantly increases classification accuracy in multilingual surveillance environments.

After language identification, the system performs real-time machine translation using MarianMT neural translation models, with automatic fallback to Google Translate whenever a dedicated translation model is unavailable.

9.2 Language Processing Architecture

The proposed language processing subsystem follows a multi-stage architecture designed to maximize reliability and accuracy. The workflow begins after speech transcription has been generated by Whisper large-v3. The transcribed text and acoustic language probabilities are passed to the hierarchical language classifier. Based on confidence levels, the classifier determines the most probable spoken language.

The detected language is subsequently used to:

- Enable multilingual violence detection,

- Select the optimal translation engine,
- Translate text into a target language,
- Support multilingual surveillance monitoring.

The complete workflow is:

Speech Transcription → Script Detection → Whisper Language Analysis → Text Ensemble Classification → Confidence Blending → Translation → Threat Analysis

9.2.1 Advantages of the Proposed Framework

- High multilingual recognition accuracy.
- Robust performance for short speech segments.
- Reduced confusion between similar languages.
- Real-time translation support.
- Adaptability to different surveillance environments.
- Improved multilingual threat analysis.

9.3 Hierarchical Language Detection Framework

One major contribution of this work is the implementation of a hierarchical language classifier rather than relying on a single prediction engine. Instead of making decisions based solely on text or audio, the proposed framework combines multiple information sources according to their reliability.

The language detection hierarchy consists of three levels:

- **Level 1 — Unicode Script Heuristic:** Fast and highly reliable script-based detection.
- **Level 2 — Whisper Acoustic Language Detection:** Primary language prediction based on speech acoustics.
- **Level 3 — Ensemble Text Classification:** Backup mechanism using multiple text engines.

This hierarchy improves classification accuracy while reducing misclassification in multilingual scenarios. For example, Whisper acoustic detection successfully differentiates French from Dutch, which text-only systems often confuse.

9.4 Unicode Script Heuristic

The first stage of language detection involves identifying the dominant writing system used in the transcribed text. Certain languages employ highly distinctive scripts that can be recognized almost perfectly without advanced machine learning:

Script	Language
Arabic	Arabic
Hebrew	Hebrew
Devanagari	Hindi
Hangul	Korean
Greek	Greek
Cyrillic	Russian
Chinese Characters	Chinese
Kana + Kanji	Japanese

Table 9.1: Unicode Script to Language Mapping

The implemented system counts character frequencies from Unicode ranges and determines the dominant script present in the text. The script dominance ratio is calculated as:

$$\text{Dominance} = \frac{N_{\text{script}}}{N_{\text{total}}}$$

where N_{script} represents characters belonging to a specific script and N_{total} denotes total detected characters. If script dominance exceeds 60%, the language is classified immediately. This method provides near-perfect performance for non-Latin languages because scripts such as Arabic or Korean are unique to specific linguistic families.

9.4.1 Benefits of Script-Based Detection

- Extremely fast execution.
- High reliability.
- Minimal computational cost.
- Immediate language recognition for non-Latin scripts.

9.5 Whisper-Based Language Detection

After script detection, the second stage employs Whisper large-v3 acoustic language detection, which represents the primary language identification mechanism. Unlike conventional methods that classify languages from text, Whisper analyzes audio spectrograms directly, exploiting acoustic characteristics such as pronunciation style, phonetic structure, acoustic rhythm, and language-specific speech patterns.

The Whisper model returns a probability distribution across 99 supported languages. The most probable language is selected:

$$L^* = \arg \max_L P(L \mid A)$$

where L^* is the predicted language and $P(L | A)$ is the probability of language L given audio signal A .

9.5.1 Confidence Threshold Strategy

The proposed implementation introduces two confidence thresholds:

Threshold	Value
High Confidence	0.85
Low Confidence	0.40

Table 9.2: Whisper Confidence Thresholds

When Whisper confidence exceeds 0.85, the system trusts Whisper completely. If confidence falls between 0.40 and 0.85, Whisper is blended with text engines. When confidence is below 0.40, text engines dominate classification.

9.6 Text-Based Ensemble Classification

Although Whisper provides highly accurate language detection, some situations may still produce uncertainty, particularly in short speech segments. To improve robustness, the system integrates multiple text classification engines.

9.6.1 LangDetect Engine

The langdetect library provides probabilistic language classification based on statistical text analysis. Its prediction confidence contributes significantly to final classification.

9.6.2 LangID Engine

The langid classifier offers language identification optimized for short texts. Its confidence values are normalized to ensure compatibility with other engines.

9.6.3 Trigram Language Heuristics

A lightweight trigram scoring mechanism was implemented to improve recognition reliability. Each language contains common linguistic patterns (e.g., English: *the, ing, tion*; French: *les, eau, aux*; German: *sch, ung, ich*). The trigram score is calculated as:

$$\text{Score}(L) = \frac{M_L}{T_L}$$

where M_L represents matched trigrams and T_L denotes total language trigrams.

9.6.4 Weighted Soft Voting

The system combines predictions using weighted soft voting:

Engine	Weight	Notes
LangDetect	0.45	Statistical n-gram model; seeded for determinism
LangID	0.40	Naive Bayes classifier; fast and reliable
Trigram Heuristic	0.15	Custom per-language trigram hints

Table 9.3: Text Engine Weights in Soft Vote Mode

The combined probability is:

$$P(L) = \frac{\sum_i w_i P_i(L)}{\sum_i w_i}$$

where w_i is engine weight and $P_i(L)$ is engine probability. This ensemble approach improves stability and reduces classification uncertainty.

9.7 Confidence Blending and Temporal Smoothing

When Whisper confidence lies in the intermediate range, the system blends Whisper predictions with text-based probabilities:

$$P_{\text{blend}} = w_w P_w + w_t P_t$$

where P_w represents Whisper probability, P_t represents text-engine probability, w_w is the Whisper weight, and w_t is the text weight. The Whisper contribution dynamically increases as confidence rises.

To prevent unstable language switching, the system applies history-based temporal smoothing. Recent predictions are stored and accumulated:

$$L_t = \arg \max \sum_{i=1}^n \text{Confidence}_i$$

This smoothing mechanism improves consistency during continuous speech monitoring.

9.8 Real-Time Translation Using MarianMT

Following language detection, the system performs neural machine translation using MarianMT, an open-source neural translation framework developed by Helsinki-NLP.

9.8.1 Architecture: MarianMT + Google Fallback

Translation follows a two-tier approach:

1. **MarianMT (Primary):** Helsinki-NLP’s MarianMT models are loaded on demand and cached for subsequent calls. Inference uses beam search with 5 beams and a maximum output length of 512 tokens.
2. **Google Translate (Fallback):** When no MarianMT model exists for the required language pair, Google Translate via the `deep-translator` library is used as a reliable cloud fallback.

9.8.2 Supported MarianMT Language Pairs

Source	Target	Model
French	English	Helsinki-NLP/opus-mt-fr-en
Arabic	English	Helsinki-NLP/opus-mt-ar-en
German	English	Helsinki-NLP/opus-mt-de-en
Spanish	English	Helsinki-NLP/opus-mt-es-en
Russian	English	Helsinki-NLP/opus-mt-ru-en
Chinese	English	Helsinki-NLP/opus-mt-zh-en
Japanese	English	Helsinki-NLP/opus-mt-ja-en
English	Arabic	Helsinki-NLP/opus-mt-en-ar
English	French	Helsinki-NLP/opus-mt-en-fr
French	Arabic	Helsinki-NLP/opus-mt-fr-ar

Table 9.4: Selected Supported MarianMT Language Pairs

9.8.3 Language Code Normalization

Before translation, language codes are normalized to handle aliases: `zh-cn` $\rightarrow$ `zh`, `jap` $\rightarrow$ `ja`, `por` $\rightarrow$ `pt`, `deu` $\rightarrow$ `de`, `fra` $\rightarrow$ `fr`, etc. This ensures consistent model selection regardless of the input code format.

9.8.4 Passthrough Mode

When the detected source language matches the target translation language, no translation is performed and the original text is returned directly as the output with engine label `passthrough`.

9.9 Right-to-Left Language Handling

Special formatting challenges arise for right-to-left (RTL) languages such as Arabic, Hebrew, Persian, and Urdu. To address text display issues, the system inserts Unicode RTL markers around transcription output, ensuring correct visual rendering during terminal display and reporting. The implemented solution significantly improves readability for multilingual surveillance operators.

9.10 Integration with the Full Pipeline

The language classifier and translator are invoked sequentially after Whisper transcription:

1. Whisper’s `language_probs` dictionary is passed directly to `LanguageClassifier.classify()`.
2. The classifier returns a result dict including `language_code`, `confidence`, `method`, and `engines_used`.
3. If translation is enabled, `RealTimeTranslator.translate()` is called with the detected language code.

4. The detected language code is forwarded to the violence detector (Chapter 10) for language-appropriate keyword selection.

9.11 Mathematical Formulation Summary

Script Dominance

$$\text{Dominance} = \frac{N_{\text{script}}}{N_{\text{total}}}$$

Whisper Language Prediction

$$L^* = \arg \max_L P(L \mid A)$$

Ensemble Classification

$$P(L) = \frac{\sum_i w_i P_i(L)}{\sum_i w_i}$$

Confidence Blending

$$P_{\text{blend}} = w_w P_w + w_t P_t$$

Temporal Smoothing

$$L_t = \arg \max \sum_{i=1}^n \text{Confidence}_i$$

9.12 Summary

This chapter presented the multilingual language detection and real-time translation subsystem. The proposed implementation introduced a hierarchical Whisper-first classification strategy combining Unicode script analysis, acoustic language detection, and ensemble text classification. The system leveraged Whisper large-v3 as the primary language detector while integrating langdetect, langid, and trigram heuristics for improved robustness. Confidence-based blending and temporal smoothing further enhanced prediction stability. Furthermore, MarianMT neural translation models were integrated to provide multilingual translation support, while a Google Translate fallback mechanism ensured continuous operation across unsupported language pairs. The multilingual capabilities presented in this chapter provide the foundation for semantic speech understanding and multilingual threat analysis, which are explored in the following chapter.

Chapter 10

Violence Keyword Detection and Alert System

10.1 Introduction

Modern intelligent surveillance systems must go beyond passive monitoring and provide proactive threat detection capabilities. Traditional surveillance systems primarily focus on recording incidents after they occur, which limits their ability to prevent dangerous situations. Consequently, intelligent monitoring frameworks increasingly require automated mechanisms capable of identifying suspicious behavior and generating immediate alerts.

Speech content represents a valuable source of contextual information in surveillance environments because verbal communication often precedes violent incidents, threats, or criminal activities. Words associated with aggression, weapons, violence, terrorism, or physical harm may indicate elevated risk levels that require immediate attention.

To address this challenge, the proposed surveillance system incorporates an intelligent multilingual violence detection and alert subsystem capable of analyzing transcribed speech content in real time. The subsystem automatically identifies violence-related keywords across multiple languages and generates security alerts whenever suspicious speech is detected. Unlike traditional keyword matching systems limited to English, the proposed implementation supports multilingual violence detection across more than thirty languages, enabling threat analysis in globally diverse environments.

10.2 Violence Detection System Architecture

The proposed violence detection module follows a layered processing architecture integrated into the overall audio surveillance pipeline. After speech transcription and language classification have been completed, the transcribed text is forwarded to the violence detection subsystem. The module first determines the detected language and dynamically loads corresponding violence-related keywords. The system then scans the transcription using language-aware matching techniques.

The violence detection workflow can be summarized as:

Speech Recognition → Language Detection → Keyword Selection → Threat Analysis
→ Alert Generation → Security Notification

10.2.1 Advantages of the Proposed Design

- Real-time threat identification.
- Multilingual speech monitoring.
- Immediate notification generation.
- Language-aware keyword matching.
- Scalability for future languages.
- Low computational overhead.

10.3 Multilingual Violence Detection Framework

A major limitation of traditional surveillance systems is language dependency. Many conventional systems rely exclusively on English keywords, making them ineffective in multilingual environments. To overcome this issue, the proposed implementation introduces a multilingual violence keyword framework supporting over thirty languages.

Language	ISO Code	Language	ISO Code
English	en	Turkish	tr
Arabic	ar	Greek	el
French	fr	Persian	fa
German	de	Bengali	bn
Spanish	es	Vietnamese	vi
Italian	it	Thai	th
Portuguese	pt	Hebrew	he
Russian	ru	Ukrainian	uk
Chinese	zh	Polish	pl
Japanese	ja	Swedish	sv
Korean	ko	Danish	da
Hindi	hi	Norwegian	no

Table 10.1: Supported Languages for Violence Keyword Detection

Each language contains a predefined set of violence-related expressions covering physical violence, weapons, terrorism, explosions, murder, threats, bloodshed, and armed conflict. The English baseline includes 13 core threat-related terms:

kill, attack, bomb, gun, shoot, knife, fight, terror, explosion, murder,
dead, blood, threat

10.4 Language-Aware Keyword Selection

After language detection, the system dynamically selects violence keywords corresponding to the detected language. Instead of scanning every language simultaneously, the

implementation optimizes efficiency by selecting the English baseline keywords and the keywords of the detected language:

$$K_{\text{final}} = K_{\text{English}} \cup K_{\text{Language}}$$

where K_{English} denotes English baseline keywords and K_{Language} represents language-specific keywords. The union of both keyword sets improves robustness because some multilingual conversations may include mixed-language speech.

10.5 Threat Detection Methodology

The implemented threat detection mechanism analyzes transcribed speech and searches for suspicious terms indicating potential violence. The system uses two different matching approaches depending on language characteristics.

10.5.1 Latin Script Languages

For Latin-based languages such as English, French, German, and Spanish, the system employs whole-word regular expression matching:

```
re.search(r"\b{keyword}\b", text.lower())
```

Word-boundary matching ensures accurate detection while reducing false positives. For example, the keyword “gun” would match “He has a gun” but not “gunning”.

10.5.2 Non-Latin Script Languages

Languages such as Arabic, Chinese, Japanese, Korean, Hebrew, and Persian do not always rely on conventional whitespace boundaries. Therefore, regular expressions based on word boundaries become unreliable. To solve this issue, the system uses substring matching for non-Latin scripts:

$$\text{Threat} = \begin{cases} 1, & \text{Keyword} \in \text{Text} \\ 0, & \text{Otherwise} \end{cases}$$

This adaptive strategy ensures reliable multilingual detection.

10.6 Threat Detection Workflow

The threat analysis workflow consists of the following sequential stages:

1. **Audio Capture:** Audio is captured from surveillance microphones.
2. **Speech Transcription:** Whisper large-v3 converts speech into text.
3. **Language Identification:** The language classifier determines the spoken language.
4. **Keyword Retrieval:** Language-specific violence keywords are loaded.
5. **Text Scanning:** Speech content is scanned for suspicious keywords.

6. **Threat Decision:** If suspicious words are found, a threat is flagged; otherwise, the conversation is considered safe.
7. **Alert Transmission:** Detected threats trigger an automated email notification.

10.7 Email-Based Security Alert System

One of the key contributions of this work is the implementation of an automated alert generation mechanism. Whenever suspicious speech is detected, the system immediately sends a formatted email notification to designated recipients.

10.7.1 Alert Trigger

An alert is triggered when matched keywords are found. The alert contains:

- Timestamp of detection.
- Identified speaker name (from ECAPA-TDNN).
- All matched keywords (highlighted in red).
- Full transcription with matched keywords visually highlighted.
- Translation of the transcription (if translation was enabled).

10.7.2 SMTP Configuration

Parameter	Value/Source
SMTP Host	smtp.gmail.com
SMTP Port	587 (STARTTLS)
Sender Email	Environment variable <code>ALERT_SENDER_EMAIL</code>
Sender Password	Environment variable <code>ALERT_SENDER_PASSWORD</code>
Recipient Email	Configurable constant

Table 10.2: SMTP Alert Configuration

Credentials are read from environment variables rather than hardcoded in source code, following security best practices.

10.8 HTML Alert Formatting

To improve readability, the system generates visually enhanced HTML emails. The email design includes red warning banners, highlighted threat keywords, structured tables, clear transcription display, and an optional translation section. Threat words are highlighted using red background with white text, helping operators rapidly identify dangerous content without reading entire messages.

The alert email is composed in both plain-text and HTML formats:

- The plain-text part ensures delivery to clients that do not render HTML.
- The HTML part uses inline CSS for a professional layout with a red alert header, a metadata table, highlighted keywords in red badges, and the transcription with matched keywords marked in a red-background highlight span.

10.9 Right-to-Left Language Support

Right-to-left languages such as Arabic and Hebrew introduce formatting challenges during terminal display and alert rendering. To solve this issue, the system inserts Unicode Right-to-Left markers around detected keywords. This ensures correct visualization for Arabic threat words and improves multilingual usability and operator experience.

10.10 Full Pipeline Integration

The `AudioProcessingPipeline.run()` method orchestrates all audio modules in the following sequence:

1. **Audio Capture:** Record fixed-duration or VAD-triggered audio.
2. **Speaker Recognition:** Identify the speaker using ECAPA-TDNN.
3. **Speech-to-Text:** Transcribe with Whisper large-v3 and extract `language_probs`.
4. **Language Classification:** Apply the Whisper-first hierarchical classifier.
5. **Translation (optional):** Translate using MarianMT or Google Translate.
6. **Violence Keyword Scan:** Scan transcription with the detected language's keyword list.
7. **Alert Dispatch:** Send email if keywords found.
8. **Cleanup:** Delete temporary WAV file.

10.11 Command-Line Interface

The complete pipeline is accessible through a comprehensive CLI. Key options include:

```
1 # Run full pipeline (record + transcribe + detect + alert)
2 python pipeline.py
3
4 # Transcribe an existing WAV file
5 python pipeline.py --file audio.wav --translate en
6
7 # Enroll a speaker
8 python pipeline.py --enroll "Alice" alice1.wav alice2.wav
9
10 # List enrolled speakers
11 python pipeline.py --list-speakers
```

```

12
13 # Use a specific Whisper model size
14 python pipeline.py --whisper-model medium
15
16 # Use Azure Speech backend
17 python pipeline.py --backend azure --azure-key KEY --azure-region
    eastus
18
19 # Disable speaker identification and violence check
20 python pipeline.py --no-speaker-id --no-violence-check

```

10.12 Example Pipeline Output

A typical pipeline execution produces output similar to the following:

```

1 Recording for 5s ...
2 Transcribing with Whisper large-v3 ...
3   Whisper language detection: ar (94.3% confidence)
4   Top-5: ar=94.3%  fa=2.1%  ur=1.2%  he=0.8%  en=0.6%
5
6 Transcription: [Arabic RTL text]
7
8 Language: Arabic (ar)  conf=98%  method=[whisper_primary]
9
10 Speaker: Alice [accepted, cos=0.82]
11
12 Translation (English): "I will kill the target tonight"
13
14 THREAT DETECTED -- keywords: [kill, target]
15   Alert sent to security@example.com

```

10.13 Mathematical Formulation Summary

Keyword Union

$$K_{\text{final}} = K_{\text{English}} \cup K_{\text{Language}}$$

Threat Decision Function

$$\text{Threat} = \begin{cases} 1, & \text{Keyword} \in \text{Text} \\ 0, & \text{Otherwise} \end{cases}$$

Threat Probability

$$P(\text{Threat}) = \frac{N_{\text{matched}}}{N_{\text{keywords}}}$$

where N_{matched} represents detected keywords and N_{keywords} represents total monitored keywords. Higher values indicate greater threat severity.

10.14 Limitations and Future Improvements

Despite strong performance, the proposed violence detection framework has several limitations. Keyword-based detection may occasionally generate false positives when violent words are used in harmless contexts (e.g., “This movie contains violence.”). Sarcasm and contextual meaning also remain difficult to interpret.

Future improvements may include:

- **Context-Aware NLP:** Replace keyword matching with sentence-level intent classification using a fine-tuned multilingual transformer (mBERT or XLM-RoBERTa) to reduce false positives.
- **Expanded Language Coverage:** Add keyword lists for additional languages as the system is deployed in new regions.
- **Risk Scoring:** Assign graduated risk scores based on keyword severity, number of matches, and speaker identity to prioritize alerts.
- **Audio-Visual Fusion:** Combine audio threat detection with video-based violence detection for higher confidence alerting.
- **Streaming Mode:** Implement true real-time streaming transcription using Whisper’s chunked processing API to reduce latency.
- **Offline Alert Fallback:** Store undelivered alerts locally and retry when network connectivity is restored.

10.15 Summary

This chapter presented the multilingual violence detection and alert subsystem. The proposed implementation introduced a language-aware threat detection mechanism capable of identifying dangerous speech across more than thirty languages. The subsystem dynamically selects violence-related keywords according to detected language and applies adaptive matching strategies for Latin and non-Latin scripts. Furthermore, an automated email alert mechanism was integrated to notify surveillance operators whenever suspicious speech is detected. The implemented framework transforms the surveillance system from passive monitoring into a proactive security intelligence solution capable of identifying verbal threats in real time, thereby improving safety and situational awareness in multilingual environments. This completes the three-chapter coverage of the audio processing pipeline.

Part IV

Database, Ethics, and Conclusion

Chapter 11

Database and System Integration

11.1 Introduction

This chapter describes the database design that underlies the surveillance system's visitor management, retrospective search, and forensic investigation capabilities. We also describe how the visual and audio modules are integrated into a unified operational pipeline.

11.2 Data Storage Strategy

The database maintains the following record structure for each detected individual:

- Unique visitor identifier (UUID).
- Facial embedding (128-dimensional float vector).
- Timestamp of first visit.
- Timestamp of most recent visit.
- Estimated age range.
- Estimated gender.
- Estimated ethnicity.
- Detected emotion at time of capture.
- Glasses status (binary).
- Facial hair status (binary).
- Visit count.
- Camera locations where the visitor was detected.

11.3 Similarity Search

The database supports similarity searches using cosine similarity between facial embeddings. When a new face is detected, its embedding is compared against all stored embeddings. A predefined threshold determines whether the match confidence is sufficient to classify the person as a returning visitor. This enables visitor re-identification across multiple visits and camera locations without storing raw facial images.

11.4 Backward Identification

For security investigations, the database enables filtering based on demographic attributes. Security personnel can input known characteristics (e.g., male, 20s, wearing glasses, Asian) and the system returns a candidate list of visitors matching those criteria within a specified time range. This human-in-the-loop approach assists investigations while maintaining appropriate oversight and reducing manual review time.

11.5 Retrospective Identification Use Case

In the event of a reported incident such as theft, security personnel provide a reference image of the suspect. The system encodes this image and performs a database-wide search to identify matching or visually similar individuals. When an exact identity match is unavailable, demographic attributes (age range, gender, ethnicity) narrow down potential candidates, significantly reducing investigation time. The database maintains privacy-aware storage through feature-based encoding rather than raw image retention.

11.6 System Integration

The complete system integrates visual and audio modules through a shared event log and alert database. Visual detections (face recognition events, demographic estimates, violence detections) and audio detections (transcriptions, language classifications, keyword matches, speaker identifications) are timestamped and stored together, enabling investigators to correlate audio and visual evidence from the same time window.

11.7 Summary

This chapter described the database structure, similarity search mechanism, and retrospective identification workflow. In the next chapter, we discuss ethical considerations. The final chapter presents the conclusion and future work.

Chapter 12

Ethical Considerations

12.1 Introduction

The proposed system is designed to comply with ethical guidelines for responsible AI usage and does not violate ethical or legal principles when deployed correctly. Ethical considerations were integrated at both the design and implementation stages across all modules.

12.2 Purpose Limitation

The system is intended solely for security assistance, visitor analytics, and operational support within malls or public spaces. It is not designed for law enforcement, legal judgment, or autonomous decision-making. The scope of use is clearly defined to prevent mission creep into inappropriate applications.

12.3 Human-in-the-Loop Design

The system does not take actions independently. All outputs (identity matches, age range, gender, appearance-based attributes, transcriptions, keyword alerts) are advisory. Final decisions are always made by human security personnel through manual review. This design ensures that technology enhances rather than replaces human judgment.

12.4 Use of Soft Biometrics

Attributes such as age, gender, and ethnicity are treated as soft biometric features. These attributes are probabilistic estimates, not definitive facts. They are used only for filtering and narrowing down candidates, never for accusation or profiling. The system explicitly acknowledges the limitations of demographic estimation.

12.5 Privacy Preservation

Facial data is stored as numerical embeddings rather than raw facial images. Embeddings cannot be reverse-engineered to reconstruct the original face. Audio data is processed

transiently and not retained beyond the processing window unless a threat is detected, in which case only the transcription and metadata are stored. Data access is restricted to authorized personnel only.

12.6 Bias Awareness and Mitigation

The system acknowledges the presence of potential bias in demographic estimation models. Diverse datasets (FairFace, RWF-2000) were prioritized during training to minimize bias. Outputs are never used in isolation and are always cross-checked by humans.

12.6.1 Audio Bias

Speech recognition and language detection models may exhibit performance variation across accents, dialects, or languages. Mitigation strategies include using high-accuracy multilingual models (Whisper large-v3, ECAPA-TDNN), allowing future expansion to support additional languages and dialects, and avoiding decision-making based solely on automated outputs without human review.

12.6.2 Ethical Use in Security Applications

Future security-related features must follow strict ethical guidelines:

- The system should flag potential risks, not label individuals.
- Final decisions must remain under human supervision.
- False positives must be handled carefully to prevent unnecessary alarm.
- The goal is risk prevention, not mass surveillance.

12.7 Legal Compliance

The system must be deployed in compliance with applicable data protection regulations (e.g., GDPR in Europe, PDPA in applicable jurisdictions). Operators are responsible for obtaining necessary permissions for audio and video monitoring in their deployment environment and for informing individuals that surveillance is in operation.

12.8 Ethical Compliance Summary

In summary, the system:

- Does not perform surveillance for punishment or judgment.
- Does not identify criminals autonomously.
- Does not rely on demographic attributes alone for decision-making.
- Operates as a decision-support tool with mandatory human oversight.

- Stores biometric data in non-reversible embedding form.
- Flags audio threats for human review rather than taking autonomous action.

12.9 Summary

This chapter described the ethical framework governing the system's design and use. In the final chapter, we present the conclusion and future work.

Chapter 13

Conclusion and Future Work

13.1 Introduction

This final chapter summarizes the achievements of the AI-Based Intelligent Surveillance System project, reflects on the lessons learned during development, and outlines the roadmap for future enhancements.

13.2 Summary of Accomplishments

This graduation project has successfully developed and validated the core AI modules required for an intelligent surveillance system. The system demonstrates the feasibility of combining face recognition, demographic analysis, and audio processing to enhance security operations while maintaining ethical safeguards.

Key technical achievements include:

- **Ethnicity Detection:** 82.56% accuracy using ResNet-18 trained on FairFace.
- **Emotion Recognition:** 73% accuracy with EfficientNet architecture.
- **Age Estimation:** 77.23% accuracy using MobileNetV2 across nine age ranges.
- **Gender Classification:** 99% accuracy with the FairFace ONNX model.
- **Violence Detection (Video):** Trained on the RWF-2000 real-world surveillance dataset.
- **Glasses Detection:** 90% accuracy using ResNet-50 with transfer learning.
- **Head Pose Estimation:** MAE $\approx 3.8^\circ$ for yaw/pitch/roll using MobileNetV2.
- **Facial Hair Detection:** Binary classification using ResNet-50.
- **Face Recognition & Identity Management:** 128-dimensional embeddings with per-speaker adaptive thresholds.
- **Audio Pipeline:** Whisper large-v3 STT with Whisper-first hierarchical language classification across 99 languages.

- **Speaker Recognition:** ECAPA-TDNN with augmented enrollment and per-speaker cosine thresholds.
- **Multilingual Translation:** MarianMT (40+ language pairs) with Google Translate fallback.
- **Violence Keyword Detection:** 30+ languages with language-aware regex/substring matching.
- **Automated Alert System:** HTML email alerts via SMTP/STARTTLS with keyword highlighting.
- **Ethical Framework:** Human-in-the-loop design with privacy-preserving embedding storage.

13.3 Lessons Learned

Throughout this project, the team developed technical expertise and valuable soft skills:

- **Dataset Quality Matters:** Balanced, diverse datasets (FairFace, RWF-2000) significantly improve real-world generalization compared to skewed alternatives.
- **Ensemble Approaches Win:** Both the language classifier (Whisper + text engines) and the speaker recognizer (augmented enrollment centroids) benefit from ensemble thinking.
- **Acoustic Before Text:** Using Whisper’s acoustic language detection before text-based engines resolved systematic French/Dutch confusion that plagued text-only approaches.
- **Modular Design Pays Off:** Building each module independently with standardized interfaces enabled parallel development and straightforward integration.
- **Ethics Must Be Built In:** Retrofitting ethical constraints is harder than designing for them from the start.

13.4 Future Work

In the next development phase, future work will focus on:

13.4.1 Visual Modules

- Integrating all modules into a unified real-time pipeline with multi-camera support.
- Expanding violence detection to include theft and child kidnapping behavior analysis.
- Developing additional facial attribute modules (nose shape, lip shape).
- Conducting bias audits and fairness assessments across all demographic modules.
- Optimizing for edge deployment on resource-constrained devices.

13.4.2 Audio Module

- Replacing keyword matching with context-aware multilingual NLP classifiers (mBERT, XLM-RoBERTa) to reduce false positives.
- Implementing graduated risk scoring based on keyword severity and speaker identity.
- Deploying true real-time streaming transcription using chunked Whisper processing.
- Adding audio-visual fusion: correlating audio threat detections with simultaneous video violence detections for higher-confidence alerts.
- Expanding language keyword coverage to additional ISO 639-1 languages.
- Implementing offline alert queuing and automatic retry for network-degraded environments.

13.4.3 System Integration

- Deploying the full system in a real-world simulated environment (e.g., a controlled mall setting) for comprehensive end-to-end testing.
- Developing an analytics dashboard for business intelligence and security reporting.
- Implementing multi-language support expansion for the user interface.

13.5 Final Remarks

This work represents a practical application of artificial intelligence, computer vision, and natural language processing in real-world surveillance systems. The project demonstrates that modern deep learning approaches can provide valuable capabilities for public safety and business intelligence when implemented responsibly with appropriate ethical considerations.

The modular architecture ensures that individual components can be improved and expanded over time, positioning the system for continued evolution. The vision of AI-enhanced public spaces is achievable through careful development, rigorous testing, and unwavering commitment to ethical principles.

This graduation project provides a solid foundation for that vision while maintaining the human oversight necessary for responsible deployment. The balance between technological capability and ethical constraint defines the success of this system—ensuring that AI serves humanity while respecting fundamental rights and dignity.

References

- 1.
- 2.
- 3.
- 4.
- 5.